

# Lesson 4: Piecewise Functions

| Criteria            | Excellent (4)                                                                         | Proficient (3)                                                             | Developing (2)                                                           | Inadequate (1)                                       |
|---------------------|---------------------------------------------------------------------------------------|----------------------------------------------------------------------------|--------------------------------------------------------------------------|------------------------------------------------------|
| Function Definition | All pieces defined correctly with precise domain constraints; no overlaps or gaps.    | Most pieces defined correctly; minor domain issues (e.g., $<$ vs $\leq$ ). | Multiple errors in defining equations or domain intervals.               | Equations are incorrect or missing major components. |
| Graphing Accuracy   | All pieces graphed correctly, endpoints are precise, and correct open/closed circles. | Most of the graph is accurate; minor endpoint errors.                      | Major graphing errors; functions do not match constraints.               | Graph is inaccurate or missing.                      |
| Domain & Range      | Correct domain/range identified in interval notation for the entire function.         | Correct domain/range for most parts; minor notation errors.                | Errors in identifying domain/range for individual pieces.                | Domain/Range incorrectly identified or missing.      |
| Evaluation          | Correctly evaluates for x-values, identifying the correct interval, showing all work. | Mostly accurate evaluation; minor calculation errors.                      | Correctly identifies interval, but makes significant calculation errors. | Incorrect evaluation and no work shown.              |
| Application         | Real-world scenario modeled perfectly; units and constraints well-defined.            | Scenario is modeled well; some minor issues with constraints or units.     | Attempts scenario but lacks connection between model and context.        | Model does not represent the real-world scenario.    |